

RUBAN VISHNU PANDIAN VENKATASUBRAMANIAN

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EDUCATION

University of Texas at Austin

PhD in Electrical and Computer Engineering

Austin, United States

Aug 2025 - Present

Indian Institute of Technology Madras

B.Tech (Honours) + M.Tech, Electrical Engineering; CGPA: 9.38/10

Chennai, India

Aug 2019 - May 2024

AWARDS & ACHIEVEMENTS

- Ranked **1st in the Department of Electrical engineering** within the dual degree program.
- Awarded the **Philips India Prize** for having the best academic record in the dual degree program in Electrical engineering.
- Awarded **Institute Merit Prize** for scoring the highest CGPA in the fourth year of the dual degree program in Electrical engineering.
- Awarded branch change from the former branch of **Biological Sciences to Electrical Engineering** for securing a rank within top **10 % of the freshman batch**.
- Secured an All-India Rank of **7149** (among 1,60,000 students) in **JEE Advanced 2019**.
- Awarded the **National Talent Search Exam (NTSE) Scholarship** by the Government of India.

RESEARCH EXPERIENCE

Image generation using Edgemap-aided Compressive Framework

Nov 2023 - Jun 2024

Master's thesis, Co-Guided by: Prof. Nambirajan Seshadri, ECE, UC San Diego & Prof. David Koilpillai, EE Department, IIT Madras

- Designed image generative models with inspiration obtained from **classic information theory** and the **latent diffusion model**.
- Successfully demonstrated the **reconstructive and generative abilities** of our model to create **face images from edgemap inputs**.
- Work got published in the ICASSP 2025 conference.

Information-theoretic limits for crowdsourcing

Nov 2023 - Mar 2024

Co-Guided by: Prof. Andrew Thangaraj & Prof. Avhishek Chatterjee, EE Department, IIT Madras

- Worked on developing information-theoretic bounds for linear codes with **column sparsity constraints on their generator matrices**.
- Applications include labelling of large machine learning datasets using crowdsourcing.

DSP for Optical Coherence Tomography (OCT)

Aug 2023 - Dec 2023

Co-Guided by: Prof. David Koilpillai & Prof. Shanti Bhattacharya, EE Department, IIT Madras

- Studied the properties of optical data used in OCT for improving its frequency resolution, researched various methods such as **Synchroqueezing using Wigner-Ville distribution** and zero-padded DFT applied on window removed data signal.
- Successfully demonstrated the improvement in resolution of 2-layer tape data over traditional Fourier transform based methods.

Project member - 5G Testbed, IIT Madras

Jun 2022 - May 2023

Guided by: Prof. Radha Krishna Ganti, EE Department, IIT Madras

- Developed an **efficient channel equalizer model** for the MIMO transceiver, minimizing compute hardware required.
- Worked on the Tx analog frontend and performed MATLAB simulations of techniques such as **WOLA, DPD and CFR¹**.
- Analyzed DPD and its implementation on hardware, conducted fixed-point simulations to study the performance of CFR.

PROFESSIONAL EXPERIENCE

Systems Engineer - Texas Instruments (India) Pvt. Ltd.

Aug 2024 - Aug 2025

- Worked as a systems engineer in Texas Instruments India with my work focused on testing and validation of **signal processing algorithms using MATLAB simulations**.
- Studied various **clock data recovery (CDR) methods** and their performance metrics for wired communication based products.

Signal Processing Intern - Texas Instruments (India) Pvt. Ltd.

May 2023 - Jul 2023

- Studied the performance of the **duobinary signalling scheme** over a typical low-pass high speed serial link.
- Analyzed multiple equalization techniques for different signalling schemes, implemented them on **MATLAB** and obtained the SER² curves for different models.

Signal Processing Intern - Texas Instruments (India) Pvt. Ltd.

May 2022 - Jul 2022

- Designed digital filters, implemented them on firmware and explored different **mathematical algorithms to reduce the design complexity**.
- Explored different matrix inversion algorithms and **independently discovered a method** present in literature to invert a particularly structured matrix.
- Conducted performance analysis and comparison checks of the developed solution against standard implementations.

¹Windowed overlap and add, Digital pre-distortion and Crest factor reduction respectively

²Symbol error rate

COURSE PROJECTS

EE6112: Topics in Random Processes and Concentrations

Apr 2024

Guided by: Prof. Avhishek Chatterjee, EE Department, IIT Madras

- Problem statement: To compare the distributions involved when sampling items from a bag with and without replacement.
- My work involved studying both the distributions, conducting literature survey and developing fundamental bounds on the Kullback-Liebler (KL) divergence measure between these distributions.

EE5121: Convex Optimization [\[Poster\]](#)

Nov 2023

Guided by: Prof. Uday Khankhoje, EE Department, IIT Madras

- Presented a poster on *Signal Decomposition via Quadratic-Separable Optimization* (QSS) and performed experiments to verify quality of signal decomposed using this toolbox against the well-known CVXPY solver.
- Tested and validated it on non-convex loss functions and NERL's soiling data, achieved accurate signal recovery with just 20% of the known input signal.

EE5179: Deep Learning for Imaging [\[Report\]](#)

Nov 2023

Guided by: Prof. Kaushik Mitra, EE Department, IIT Madras

- Presented about the paper titled *Neural Collapse: A Review on Modelling Principles and Generalization*, studied how hierarchical classification using DL models work and understood how neural collapse might affect hierarchical classifiers.

EE5143: Information theory [\[Report\]](#)

May 2023

Guided by: Prof. Andrew Thangaraj, EE Department, IIT Madras

- Presented a review of the paper titled *Compressed Sensing*, understood how sparse signal recovery and certain algorithms such as *Basis pursuit* works.

EE6110: Adaptive Signal Processing [\[Report\]](#)

Nov 2022

Guided by: Prof. Arun Pachai Kannu, EE Department, IIT Madras

- Presented a review of the paper titled *A Family of Robust M-Shaped Error Weighted Least Mean Square Algorithms: Performance Analysis and Echo Cancellation Application*.
- Experimented with the proposed model in MATLAB and demonstrated performance gains provided compared to the standard LMS algorithm for a system tracking problem.

TEACHING EXPERIENCE

EE5143: Information Theory

Jan 2024 - May 2024

Prepared tutorials and solutions, conducted tutorial sessions, evaluated presentations done by students and taught a lecture on the topic of Hypothesis testing.

EE3110: Probability Foundations for Electrical Engineers

Aug 2023 - Nov 2023

Prepared tutorials on a biweekly basis, conducted tutorial sessions, evaluated assignments and quiz submissions and actively engaged in discussions with students regarding their doubts outside class hours.

EE3005: Communication Systems

Jan 2023 - May 2023

Handled a class of ~70 students, conducted tutorial sessions and prepared solutions for the same and facilitated doubt discussions off class hours (only undergraduate TA).

RELEVANT COURSEWORK

Signal processing and Machine learning: Convex Optimization, Deep Learning for imaging, Topics in Random Processes and Concentrations, Image Signal Processing, Adaptive Signal Processing, Information theory, Estimation theory, Error Control Coding, Advanced topics in Communication, Digital Communication Systems, Probability and Linear Algebra.

MOOCs: Graph theory, Multirate DSP (NPTEL), Neural Networks and Deep Learning, Machine Learning (Coursera)

POSITIONS OF RESPONSIBILITY

Strategist - Electronics Club, Centre for innovation, IIT Madras

Apr 2021 - Apr 2022

- Mentored a project undertaken by the club, conducted periodic sessions and curated resources for the same.
- Organized and contributed to the summer school sessions conducted for the freshman batch students.

Coordinator - Electronics Club, Centre for innovation, IIT Madras

Apr 2020 - Apr 2021

- Worked on a club project titled **Digital pen**, explored feasible ideas and contributed to design decisions.
- Curated resources, prepared presentations and conducted club sessions for the freshman batch students.
- Explored the domains of **Deep learning** and **Digital design** and worked on various mini-projects related to the said domains.

EXTRACURRICULARS

• Shaastra workshop - Introduction to FPGA:

Jan 2021

Conducted the workshop and contributed by teaching **Digital electronics and Verilog basics** in the institute's annual technical fest **Shaastra**.

• Associate Executive - Development and Association, E-Cell, IIT Madras

Oct 2019 - Jan 2020

Worked as an associate executive in the Development and Association vertical of the institute Entrepreneurship Cell, assisted in organizing and managing sessions in the institute entrepreneurship summit (E-summit) 2020.